为什么需要 Agentic OPD?

从整段 token matching，走向错误传播下的 step-wise reliability

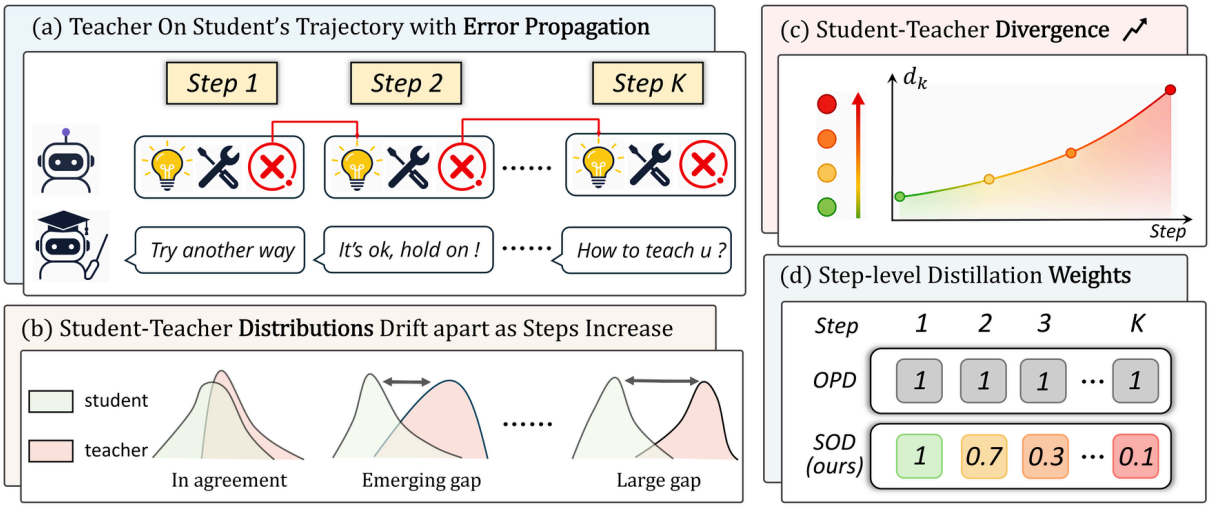
SOD

Step reliability

Step-wise On-policy Distillation for Small Language Model Agents

多轮工具交互中，一个错误 action 会注入大段异常 observation。后续状态逐步离开 Teacher support，统一累加 OPD loss 会把高方差监督当成同等可靠。

- 按模型生成步骤聚合 Student−Teacher log-prob gap，而不是把整条 trajectory 视作同质 token 流。
- 前一步分歧变大时降低当前 step 权重；策略恢复时允许权重回升，并限制最大放大幅度。
- 核心贡献是 reliability weighting；它仍以 disagreement 作为 proxy，不等价于真实因果责任。



$$d_k = \frac{1}{|\mathcal{I}_k|} \sum_{t \in \mathcal{I}_k} \left| \log \pi_\theta(y_t \mid y_{<t}) - \log \pi_{\text{teacher}}(y_t \mid y_{<t}) \right|$$

$$w_k = \min\left(w_1 \prod_{u=1}^{k-1} \frac{d_u + \epsilon}{d_{u+1} + \epsilon}, 1 + \delta \right)$$

Primary source: 2605.07725v3 · official paper figure and equations

Agentic OPD 的第一条主线：先判断当前 step 的 Teacher signal 是否还可信。

把 Hindsight 变成 Step-level Credit

StepOPSD：不改变在线 rollout，只在事后重塑 advantage

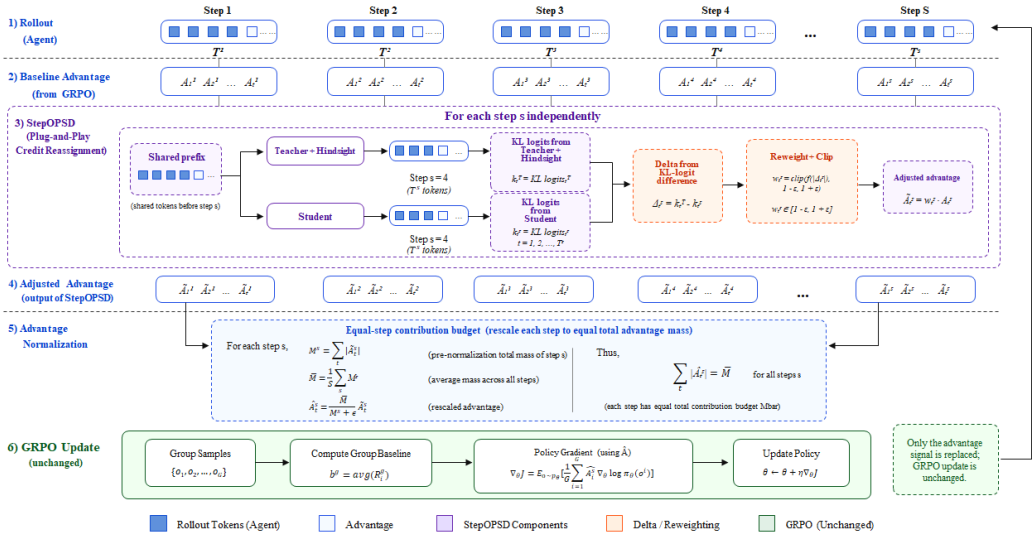
StepOPSD

Hindsight shaping

Step-Aware Online Preference Self-Distillation for Agent Reinforcement Learning

Student 仍按原策略与环境交互；rollout 完成后，将轨迹切成 causal action spans，并在加入 hindsight 的 Teacher context 下重评分同一 token。

- Teacher−Student likelihood gap 提供 dense token preference，但不替换环境给出的 outcome advantage。
- 用 advantage 符号决定 gap 的方向，再经过 sigmoid 与 clip 得到有界 credit weight。
- 最终只调节 GRPO advantage 的幅度，保留原始 outcome 的正负方向。



$$\Delta_{k,j} = \log \pi_T(z_{k,j} \mid c_k^T, z_{k,<j}) - \log \pi_S(z_{k,j} \mid c_k^S, z_{k,<j})$$

$$\tilde{A}_\ell = (1 - \lambda_{\text{mix}}) A_\ell + \lambda_{\text{mix}} (w_\ell A_\ell)$$

Primary source: 2605.27140v2 · official paper figure and equations

变化：Teacher 不再只提供 imitation target，而开始参与 outcome credit 的局部再分配。

让 OPD 感知 Turn 与 Horizon

TurnOPD：训练多深、每个 turn 占多大权重，都由在线统计控制

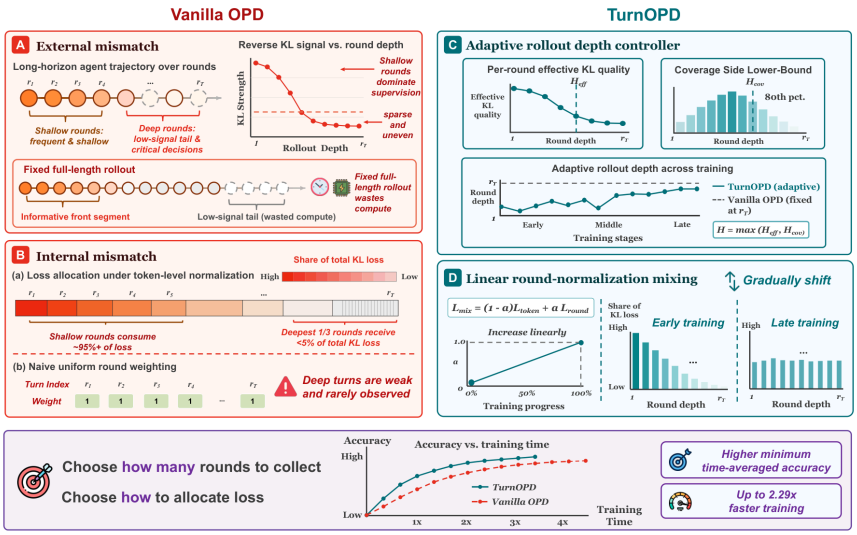
TurnOPD

Turn-aware horizon

Making On-Policy Distillation Turn-Aware for Efficient Long-Horizon Agent Training

Vanilla OPD 在长 horizon 中出现外部错配与内部错配：Teacher uncertainty 和 reverse-KL 随 turn 加深，深层 token 又可能在 trajectory normalization 下占据过多 loss。

- 用每 turn KL 与 survivor ratio 估计有效 rollout 深度。
- 再用成功轨迹长度分位数给出 coverage lower bound，避免 horizon 过短。
- 从 trajectory-normalized loss 渐进切换到 turn-normalized loss，减少长 turn 对梯度的垄断。



$$H_{\text{ctrl}} = \max(H_{\text{eff}}, H_{\text{cov}})$$

$$q_t^{\text{blend}} = (1-\alpha)\frac{n_t}{\sum_j n_j} + \alpha\frac{1}{T}$$

Primary source: 2607.05804v2 · official paper figure and equations

Agentic OPD 的第二条主线：监督不仅要可信，还要控制暴露到多深的状态分布。

控制 Student 访问的状态分布

从 DAgger 的 Teacher mixing，到 Guided-OPD 的 curriculum turn guidance

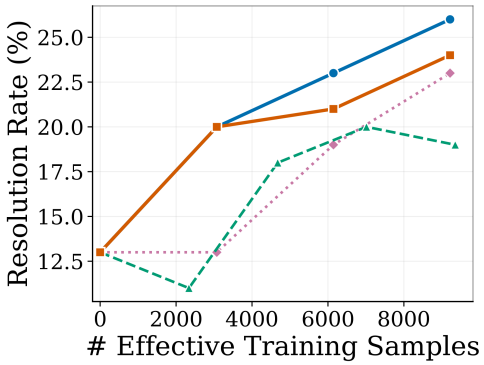
Revisiting DAgger

Mixed-policy rollout

Revisiting DAgger in the Era of LLM-Agents

仅在 Teacher demonstrations 上 SFT 会遭遇 covariate shift；纯 Student rollout 又会在冷启动时迅速进入无效状态。DAgger 在每个 turn 随机选择 Teacher 或 Student，并收集 Teacher labels。

- 逐 turn mixture 扩大可恢复状态覆盖。
- 训练目标仍是 Teacher action 的交叉熵。



$$\mu_i = \beta_i \pi_e + (1-\beta_i)\pi_{\theta_i}$$

Primary source: 2605.12913v1 · official paper figure and equations

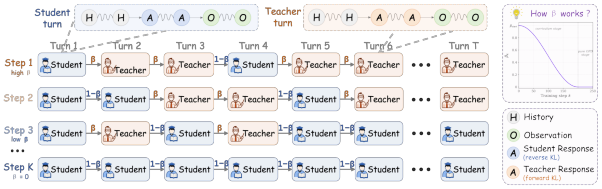
Guided-OPD

Curriculum mixing

On-Policy Distillation with Curriculum Turn-level Guidance for Multi-turn Agents

Guided-OPD 将 DAgger 式混合嵌入 OPD：训练早期 Teacher turns 较多，随后按 curriculum 衰减到纯 Student on-policy rollout。

- Teacher turn 使用 forward-KL；Student turn 使用 reverse-KL。
- 最终阶段 β=0，与标准 OPD 一致。



$$\mathcal{L}_t = (1-\beta_t)\mathcal{J}^{\pi_\theta}_{\text{rKL}} + \beta_t \mathcal{J}^{\pi_\phi}_{\text{fKL}}$$

Primary source: 2606.15912v3 · official paper figure and equations

关键取舍：Teacher intervention 改善状态覆盖，但 intervention 后的 history 不再是纯 Student on-policy。

选择性干预与 Offline Prefix Replay

不必在每个 turn 都调用 Teacher，也不必让环境在训练全程保持在线

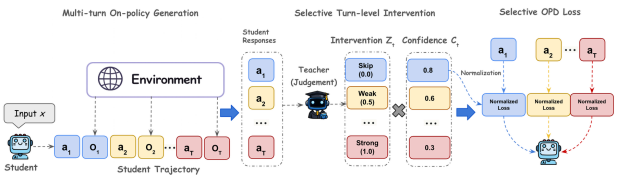
SAGE-OPD

Selective intervention

Selective Agent-Guided Intervention for Multi-Turn On-Policy Distillation

Intervention Agent 依据 Student response、Teacher disagreement 与 confidence，在 Skip / Weak / Strong 三档之间选择。

- 用 Teacher confidence 抑制不可靠干预。
- 按实际被监督 token 数归一化，保持 loss scale 稳定。



$$s_t = i_t c_t, \qquad \mathcal{L}_{\text{SAGE}} = Z \sum_t s_t \mathcal{L}_{\text{OPD}}(h_t, a_t)$$

Primary source: 2606.19659v1 · official paper figure and equations

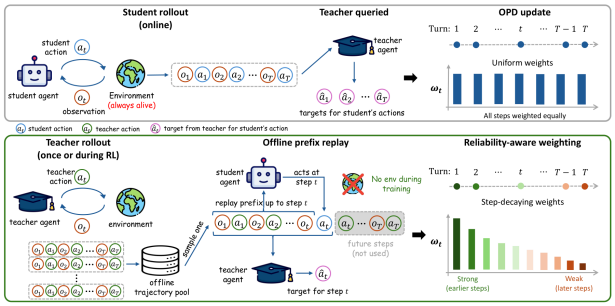
ReOPD

Offline replay

Multi-Turn On-Policy Distillation with Prefix Replay

Teacher trajectories 只需离线收集一次；Student training 通过 prefix replay 构造状态，环境无需始终在线，并让较早 step 获得更高权重。

- 用 Student / Teacher occupancy 的 geometric bridge 控制 replay distribution。
- 用指数衰减近似长期 distribution-shift cost。



$$\rho_t^*(h_t \mid x) \propto [d_t^{\theta_{\text{old}}}]^{\gamma_t} [d_t^{T}]^{1-\gamma_t}, \qquad \omega(t,\kappa) = \kappa^t$$

Primary source: 2607.04763v3 · official paper figure and equations

这一阶段开始同时优化监督质量、Teacher query 成本与 environment availability。

动态 Teacher 信息：从固定答案到 Skill

Skill-SD：Teacher privilege 来自 trajectory analysis，而不是唯一标准路径

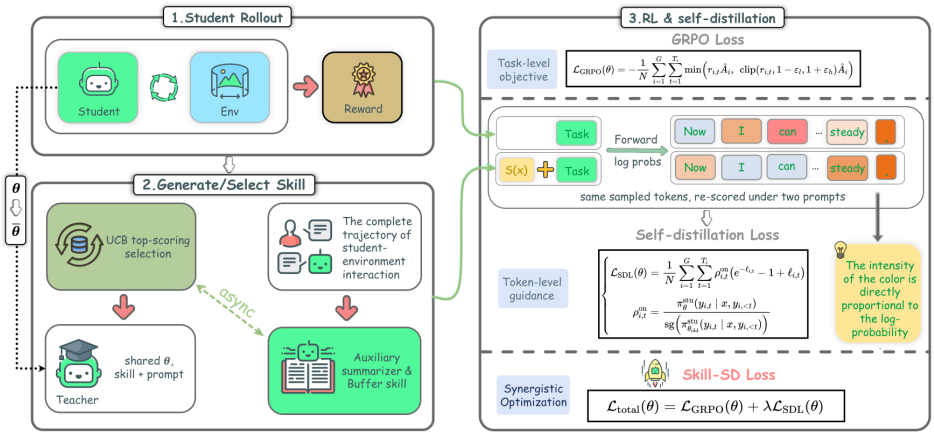
Skill-SD

Dynamic privileged context

Skill-Conditioned Self-Distillation for Multi-turn LLM Agents

Agent task 往往存在多条有效 action path。Skill-SD 将完成轨迹异步总结为 success analysis、mistake analysis 与 golden workflow，并存入 UCB-indexed skill buffer。

- Student 使用原始 prompt；Teacher 在同一 token prefix 上额外读取检索到的 skills。
- GRPO 提供 task-level credit；importance-weighted SDL 迁移 token-level Teacher distribution。
- 动态 Skill 解决 Teacher 应该知道什么，但不直接解决后续 outcome 对早期 action 的因果归因。



$$e = \mathbf{Analyze}(\tau, x) = (e_{\text{success}}, e_{\text{mistake}}, e_{\text{workflow}})$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{GRPO}} + \lambda \mathcal{L}_{\text{SDL}}$$

Primary source: 2604.10674v1 · official paper figure and equations

Agentic OPD 的第三条主线：privileged information 从静态 oracle 走向随经验演化的可检索技能。

信息不对称：Privileged Teacher 也可能误导

AR-OPD 校准 support mismatch；HINT 把 hindsight 压缩到真正的 decision fork

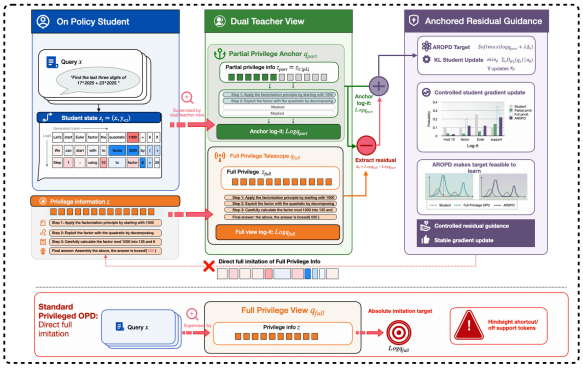
AR-OPD

Anchored residual

Beyond Absolute Imitation: Anchored Residual Guidance for Privileged On-Policy Distillation

Full privileged context 可能让 Teacher 依赖 Student 永远看不到的答案状态。AR-OPD 先构造 partial anchor，再只迁移 full view 相对 anchor 的 residual。

- Anchor 保留可由 Student 重建的支持。
- λ 控制 privileged residual 强度，避免绝对模仿。



$$\log q_\lambda = \log q_{\text{part}} + \lambda(\log q_{\text{full}} - \log q_{\text{part}}) - C_t$$

Primary source: 2606.10385v1 · official paper figure and equations

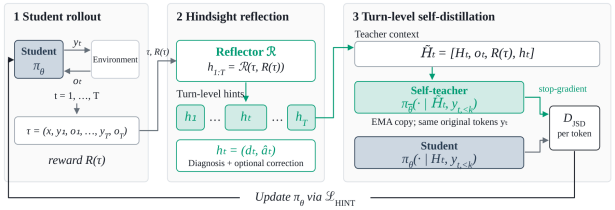
HINT

Compressed hindsight

Multi-Turn Agent Self-Distillation with Compressed Hindsight

Reflector 将完整 trajectory outcome 压缩为 turn-level diagnosis 与可选纠正动作，只在识别出的错误 turns 蒸馏。原始长图写作 HERO，但对应论文和方法实际为 HINT。

- Teacher context 注入 diagnosis，而不是泄露完整参考答案。
- JSD loss 聚焦错误 turn 的 action tokens。



$$h_t = (d_t, \hat{a}^t), \qquad \mathcal{L}_{\text{HINT}} = \frac{1}{|I(\tau)|} \sum_{t \in I(\tau)} \ell_t$$

Primary source: 2606.11559v2 · official paper figure and equations

新问题：Teacher 拥有更多信息不代表它在 Student state 上一定校准良好。

Bayesian Credit：从证据增量到 Turn Advantage

OPPO 给出 token-level value recursion；PBSD 将 privileged evidence 引入长程 Agent

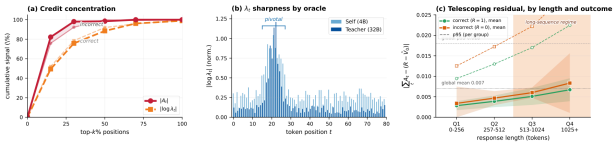
OPPO

Token Bayesian evidence

Bayesian Value Recursion for Token-Level Credit Assignment in LLM Reasoning

把每个 token 的 likelihood ratio 视为 Bayesian evidence，递归更新成功概率；相邻 value 的变化就是局部 advantage。

- Token advantages telescoping 到最终 reward 与初始 value 的差。
- 一阶近似把 credit 分解为 uncertainty × evidence。



$$V_{t+1} = \frac{\lambda_t^* V_t}{\lambda_t^* V_t + (1 - V_t)}, \qquad A_t = V_{t+1} - V_t$$

$$\sum_{t=1}^{T} A_t = R - V_0$$

Primary source: 2605.21851v2 · official paper figure and equations

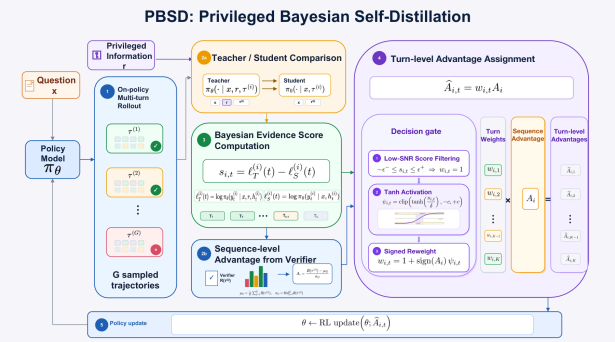
PBSD

Turn Bayesian evidence

Privileged Bayesian Self-Distillation for Long-Horizon Credit Assignment

比较有/无 privileged outcome information 时当前 action 的 likelihood，得到 turn-level evidence score，再通过 decision gate 调整 sequence advantage。

- Bayes identity 将 outcome posterior 转换为 trajectory evidence ratio。
- 保留 sequence reward 的方向，同时在 turn 内重新分配幅度。



$$s_{i,t} = \log p(a_{i,t} \mid x_i, \tau_{i,<t}, y^*) - \log p(a_{i,t} \mid x_i, \tau_{i,<t})$$

$$\tilde{A}_{i,t} = w_{i,t} A_i$$

Primary source: 2606.09348v2 · official paper figure and equations

Bayesian 视角把 Teacher gap 从“模仿误差”改写成“当前动作对成功结论提供了多少证据”。

递归与层级 Credit Assignment

AgentOPSD 用 belief revision 分 turn；CrEST 让 verifier 决定方向、Teacher 只教幅度

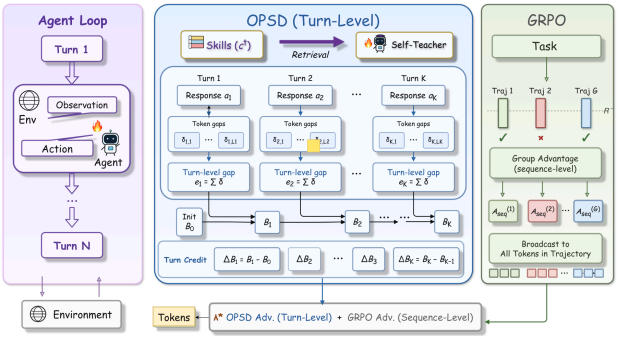
AgentOPSD

Recursive belief

Recursive Self-Distillation for Agentic Reinforcement Learning

先将 token-level privileged gap 聚合为 turn evidence，再递归更新成功 belief。相邻 belief revision 作为每个 turn 的局部 credit。

- 初始 belief 来自 group success rate。
- 最终 turn advantage 与 sequence outcome 符号对齐。



$$e_k = \log \frac{\pi_\theta(a_k \mid s_k, c^+)}{\pi_\theta(a_k \mid s_k)}, \qquad \Delta B_k = B_k - B_{k-1}$$

Primary source: 2608.05987v1 · official paper figure and equations

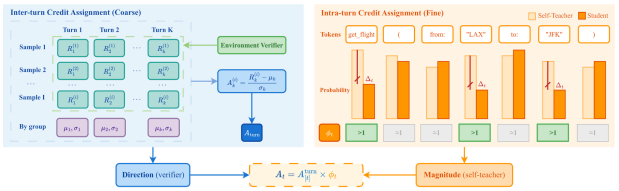
CrEST

Verifier-anchored hierarchy

Teach the Magnitude, Not the Direction

Teacher disagreement 不应决定更新方向。CrEST 让 environment verifier 提供 turn-level sign，self-teacher 只调节 token-level magnitude。

- Inter-turn credit 来自 verifier。
- Intra-turn magnitude 由 entropy-gated Teacher gap 提供。



$$\mathcal{A}_t = A^{\text{turn}}_{[t]}\phi_t, \qquad \phi_t = 1 + \lambda^{\text{eff}}_t(w_t - 1)$$

Primary source: 2608.13179v2 · official paper figure and equations

方向与幅度解耦，是减少 privileged Teacher 误导风险的一条清晰设计原则。

TASPO：Outcome-Constrained Process Supervision

用 verified successful siblings 构造 trajectory-aligned PI，同时守住 outcome advantage

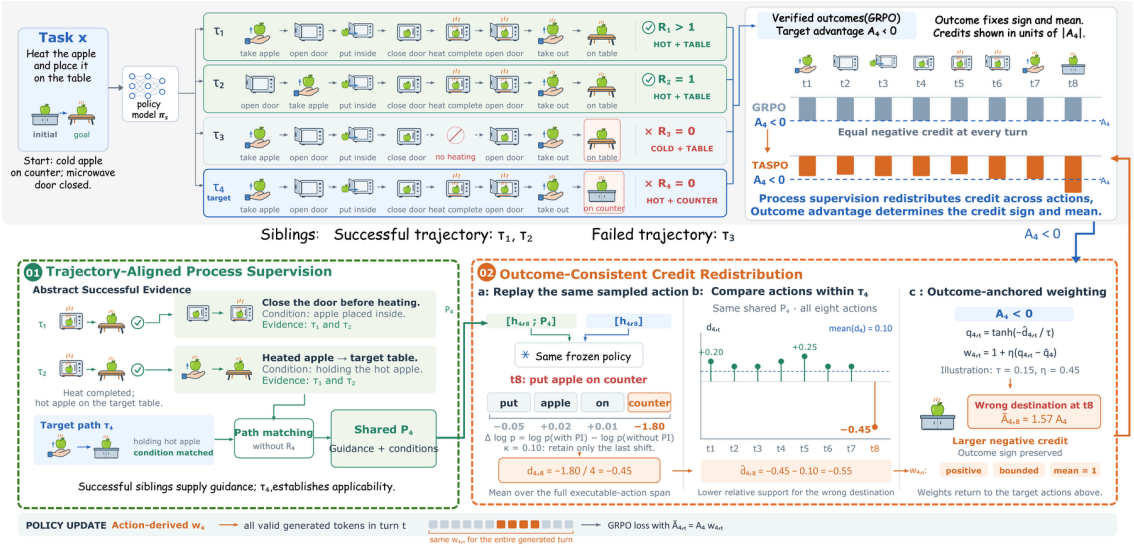
TASPO

Outcome-constrained redistribution

Reconciling Process Supervision with Outcome-Based Credit in Agentic Policy Optimization

TASPO 从同任务的 verified successful siblings 提取与当前 trajectory 对齐的 privileged information，在 action level 计算 likelihood shift，并把它转换为有界、均值为 1 的 advantage weights。

- 权重始终为正，不翻转 outcome advantage 的符号。
- 每条 trajectory 上权重均值为 1，不改变总体 advantage budget。
- 局部 process supervision 只重新分配 credit，不覆盖 verifier 给出的成败结论。



$$\tilde{A}_t^i = A_i w_t^i, \qquad 1-\epsilon_w \le w_t^i \le 1+\epsilon_w, \qquad \frac{1}{T_i}\sum_t w_t^i = 1$$

$$w_t^i = 1 + \eta(q_t^i - \bar{q}_i)$$

Primary source: 2608.31077v2 · official paper figure and equations

发展脉络：局部 KL → step reliability → state coverage → dynamic PI → Bayesian/recursive credit → outcome-constrained redistribution。